Thirty-eight Prime Outputs Refute the Ova Counting Bound
A Complete Formal Counterexample to MathDB 350342

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AI-assisted verification note

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Abstract

We verify a finite counterexample to the proposed upper bound on the number of primes produced by the Ova-Ova-Prime-Ova construction. The input primes 677 and 401 produce at least 38 distinct prime outputs. This disproves both the bound 10 in MathDB 350342 and the bound 13 in the original 2020 publication. Every displayed primality claim, residue condition, and cardinality inequality is checked in Lean. The witness was posted previously by Shivam Patel; this note claims verification, not discovery.

1 The construction and the source discrepancy

For a natural number p, write

$$k_p = p \bmod 360, \qquad f_p = \lfloor p/360 \rfloor, \qquad p = k_p + 360 f_p.$$

For two prime inputs p, q, the construction in Theorem 7 of [1] considers prime residue classes r modulo 360 for which

$$c_r = k_p + k_q + 2 - r$$

is prime. Whenever

$$v_r = r + 360(f_p + f_q)$$

is also prime, v_r is a resulting prime. Here $1 \le r < 360$ and $r \le k_p + k_q + 2$, so the subtraction is an ordinary nonnegative integer subtraction. The natural meaning of the number of resulting primes is the number of distinct such values v_r.

Conjecture 1 of the original paper [1], printed on page 118, proposes an upper bound of 13. The later arXiv paper [2] cites Theorem 7 and reuses the construction in its Goldbach discussion, but the retrieved versions do not supply the claimed bound 10. MathDB 350342 explicitly records the bound 10 [3]. These are different source statements. We do not attribute the number 10 to the original Conjecture 1. The counterexample below refutes both bounds, so this bibliographic discrepancy does not affect the mathematical outcome.

Theorem 1. *For the input primes $p = 677$ and $q = 401$, the construction has at least 38 distinct prime outputs. In particular its output count exceeds both 10 and 13.*

2 The witness and its complete verification

The inputs satisfy

$$677 = 317 + 360 \cdot 1, \qquad 401 = 41 + 360 \cdot 1.$$

Both are prime. For these inputs the two output conditions simplify to

$$c_r = 360 - r \text{ is prime}, \qquad v_r = 720 + r \text{ is prime}.$$

The following table provides 38 residues and the corresponding prime pairs.

r	$360 - r$	$720 + r$	r	$360 - r$	$720 + r$
7	353	727	221	139	941
13	347	733	233	127	953
23	337	743	247	113	967
49	311	769	251	109	971
53	307	773	257	103	977
67	293	787	263	97	983
77	283	797	271	89	991
89	271	809	277	83	997
91	269	811	289	71	1009
103	257	823	293	67	1013
109	251	829	299	61	1019
119	241	839	301	59	1021
133	227	853	313	47	1033
137	223	857	319	41	1039
161	199	881	329	31	1049
163	197	883	331	29	1051
167	193	887	341	19	1061
187	173	907	343	17	1063
209	151	929	349	11	1069

All residues lie strictly between 0 and 360. Every integer in either prime column is prime. These finite assertions can be checked by trial division using the primes at most 31, since the largest entry is 1069 and $33^2 > 1069$. Lean's numerical normalization produces checked proofs of the primality assertions rather than accepting a primality oracle.

Each residue also genuinely represents a prime residue class: its output v_r is prime and

$$v_r \bmod 360 = (720 + r) \bmod 360 = r.$$

Thus the residue-class requirement in the original construction has not been replaced with a heuristic coprimality test or omitted. The prime v_r itself is an explicit witness to that requirement.

Finally $r \mapsto 720 + r$ is injective. Since the table lists 38 different residues, it lists 38 different output primes. This proves Theorem 1. Exhaustively ruling out further outputs is unnecessary: a lower bound of 38 already contradicts either universal upper bound.

3 A precise finite-set formulation

To make the full bridge to the source explicit, let $E(p, q, r)$ mean the conjunction

$$0 < r < 360, \qquad r \le k_p + k_q + 2,\ \mathrm{Prime}(k_p + k_q + 2 - r), \qquad \mathrm{Prime}(r + 360(f_p + f_q)).$$

Define finite sets

$$R(p,q) = \{r \in \{0,\dots,359\} : E(p,q,r)\},$$
$$O(p,q) = \{r + 360(f_p + f_q) : r \in R(p,q)\}.$$

Every member of $O(p,q)$ is prime. For every $r \in R(p,q)$, its output prime is a witness that r belongs to the original set of prime residue classes. Conversely, a residue satisfying the original source's output conditions satisfies E. Hence the finite set $O(p,q)$ implements the original construction's resulting primes, rather than a larger relaxed set.

The identity

$$(k_p + k_q + 2 - r) + (r + 360(f_p + f_q)) = p + q + 2$$

holds whenever $E(p,q,r)$ holds. This confirms the source decomposition as well as the output conditions. The formal proof includes this identity for arbitrary natural inputs and every admissible residue.

The finite set W of table residues satisfies $|W| = 38$ and $W \subseteq R(677,401)$. Therefore

$$38 = |W| \le |R(677,401)| = |O(677,401)|.$$

The last equality follows from injectivity of addition of the fixed frequency term. This is the exact cardinality argument checked by Lean.

4 Formal coverage, attribution, and scope

The module `formal/OvaCount.lean` defines the input residue and frequency functions, the full numerical eligibility predicate, and the two finite sets above. Its declarations verify:

- primality of every output and the source decomposition for all inputs and admissible residues;
- equality of residue and output cardinalities, using injectivity;
- primality and residue/frequency values of the two actual inputs;
- cardinality 38 of the displayed witness set, and eligibility of every member;
- an explicit prime witness to the prime-residue-class requirement;
- existence of two prime inputs with at least 38 outputs, hence more than 10 and more than 13.

The definition `SourceEligible` explicitly conjoins the prime-residue-class condition with numerical eligibility. The equivalence `source_condition_iff` and the finite-set identity `source_outputs_eq` prove that this explicit source formulation counts the same outputs.

The final theorem `OvaCount.prime_count_counterexample` uses this source-formulated output set. A finite universal statement over the witness set is proved by splitting the 38 possible residues and checking the arithmetic and primality goals. The small finite-set cardinality is proved by kernel reduction. No unchecked native computation, admitted claim, or extra mathematical axiom is used.

The pinned Lean 4.34.1 compiler checked all 20 named definitions and results, plus the local decidability instance, with warnings as errors. The complete axiom audit uses only propositional extensionality, classical choice, and quotient soundness. All dependencies, proof hashes, and source hashes are recorded alongside the module. The manuscript and source interpretation are covered by the independent review accompanying the release.

The same inputs and the same 38 residues were already supplied by Shivam Patel in the MathDB response dated August 20, 2026. That prior response claims an exact total of 38. Our formal theorem deliberately asserts only the lower bound needed for the full disproof. The lower bound is exact enough to resolve the conjecture; it is not a partial result on a still-open case. We claim neither the witness's discovery nor a proof or disproof of the source's separate general at-least-one-output theorem. No Goldbach assertion is used anywhere in this argument.

References

[1] Y. A. Acevedo Agudelo, *A complete classification of the Mersenne's primes and its implications for computing*, Revista Politécnica 16(32) (2020), 111–119, Theorem 7 and Conjecture 1. https://doi.org/10.33571/rpolitec.v16n32a10.

[2] Y. A. Acevedo Agudelo, *Prime numbers. An alternative study using ova-angular rotations*, arXiv:2104.04522v1 and v2 (2021), discussion citing Theorem 7 of the preceding reference. https://arxiv.org/abs/2104.04522.

[3] MathDB, post 350342, *The upper-bound conjecture for the Ova-Ova-Prime-Ova construction*; counterexample by Shivam Patel, August 20, 2026. https://mathdb.com/p/350342.